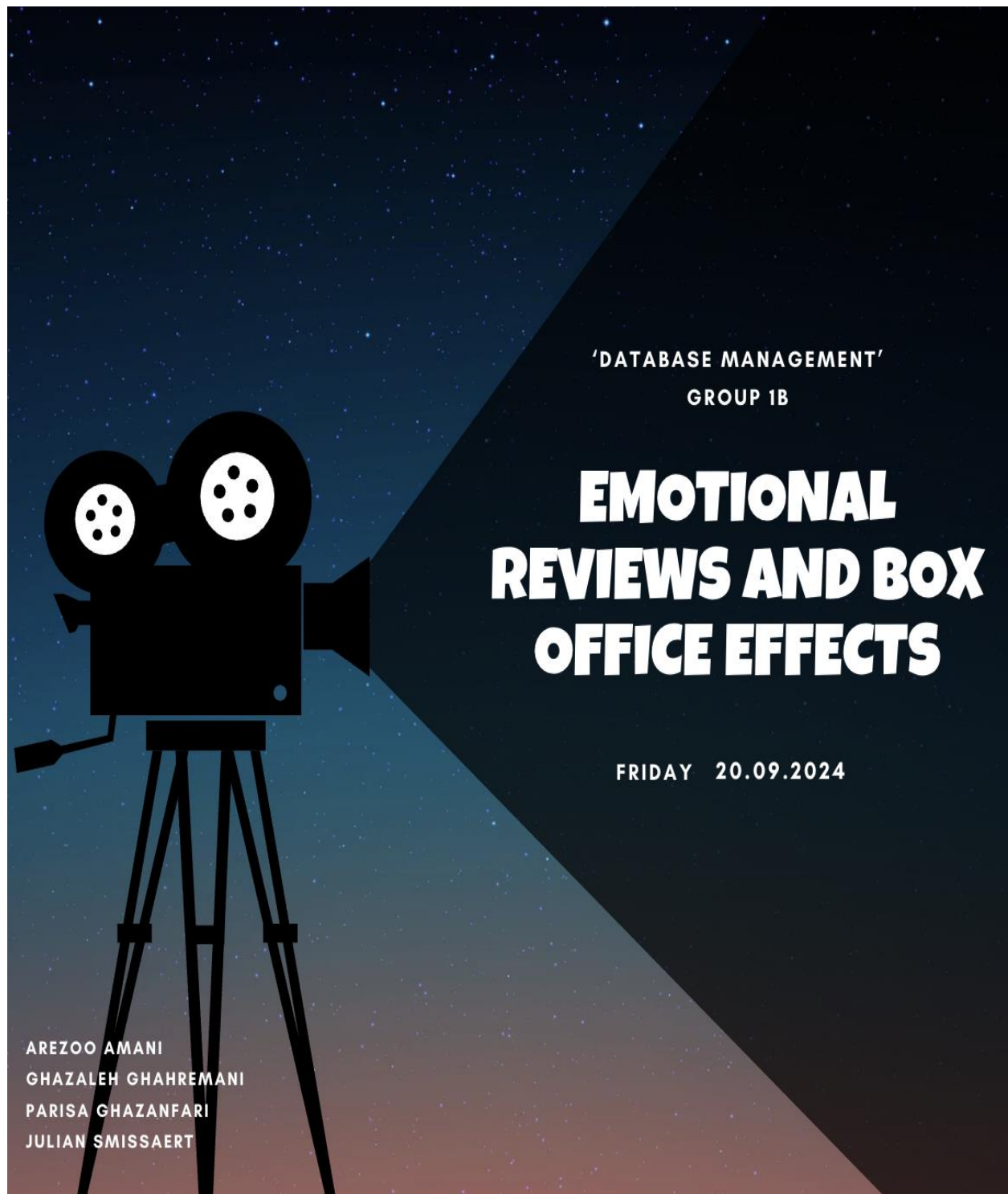




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Chapter 1 - Introduction

The movie industry is one of the largest global entertainment sectors, with box office sales serving as a critical measure of a film's success. As competition intensifies and digital platforms like IMDb, Rotten Tomatoes and Metacritic grow in influence, it is essential to understand the factors that drive audience decisions. One key element is the emotional tone expressed in movie reviews. Whether positive, negative or neutral, the emotions conveyed in reviews can shape how potential viewers perceive a film. This perception directly influences the film's box office performance.

In this project, we were given the freedom to choose our own research focus within the broad context of understanding box office determinants. We have chosen to focus on the emotional dynamics within user and expert reviews. Our goal is to explore how emotional tones in user and expert reviews influence box office performance. Emotional responses often carry weight in how audiences interpret films and understanding this relationship offers both theoretical and practical insights.

Analytical approach

For this analysis, we use four datasets, each providing important insights into different aspects of a movie's performance. The meta clean dataset offers general information about movies, helping to link data from various sources. The sales data provides financial details, such as box office earnings, production budgets and movie durations. Additionally, the user review data includes user-generated reviews with scores and feedback, along with metrics like review length and emotional tone (positive or negative). Finally, the expert review data includes expert reviews along with scores, text length, sentence structure and emotional tone.

Using these datasets, we conduct an in-depth analysis to examine how emotional tone in user and expert reviews is linked to box office success. By considering different genres, we can also explore how emotional dynamics vary across genres, offering a more detailed understanding of their impact on a film's commercial performance.

Practical impact and broader applications

Understanding the emotional dynamics in reviews offers significant practical value for film studios and marketers. Insights from this research can help studios optimize their marketing strategies by identifying which type of reviews (user or expert) has the most influence on audience decision-making. Although this research is focused on the movie industry, understanding how emotional content in reviews influences consumer behavior could provide insights relevant to other sectors. Industries such as hospitality and retail also rely heavily on customer reviews and the patterns uncovered here may inspire further investigation into how emotions in reviews impact purchasing decisions in these fields.

1.1 Main Question:

How do emotional dynamics in reviews influence the overall box office performance of movies?

1.2 Sub-questions:

1. How does the emotional tone (positive or negative) in user reviews impact the box office performance of movies across different genres?
2. What is the influence of user and expert review ratings on box office performance during the opening weekend?
3. How do positive expert reviews influence the overall box office performance of movies?
4. How do larger production budgets influence the box office performance of movies that receive negative reviews?
5. How do negative emotions in expert reviews affect the box office performance across different film genres

The choice of these main and sub-questions is aimed at gaining a better understanding of how emotions in reviews affect a film's box office performance. By examining different genres, the opening weekend and production budgets, patterns and differences in how reviews impact movie sales can be uncovered. This approach reveals not only the direct effects of reviews but also the factors that might amplify or reduce their impact.

1.3 Hypotheses

Based on literature review and available data, hypotheses have been formulated to investigate the impact of emotional dynamics in reviews on the box office performance of films. These hypotheses have been developed to test the assumptions derived from previous research and to further explore the questions posed in the study.

Key metrics and definitions for hypothesis testing are provided to ensure clarity and precision in evaluating the relationships and impacts. This includes detailed explanations of the measurements used to assess the emotional dynamics in reviews and their correlation with box office performance.

Hypothesis 1:

The relationship between emotional tone in user reviews and box office performance will vary across genres.

Emotional tone in user reviews:

The analysis of sentiment will be employed to assess the emotional tone of user feedback. A score for the emotional tone of each review will be given, ranging from 0% (totally negative) to 100% (totally favorable). "Higher emotional tone" is defined for this hypothesis as an average emotional tone score (e.g., >50%) over the median value across all user evaluations for a particular genre.

Hypothesis 2:

Movies with higher user and expert reviews ratings will have a stronger opening weekend box office performance.

Higher user and expert review ratings:

Defined as films with an average expert review score above the median score of all expert reviews in the dataset (e.g., a score > 6 on a scale of 1-10 for expert review, and score > 60 on a scale of 1-100 for user review.)

Hypothesis 3:

Movies with more positive expert reviews will perform better at the box office than movies with fewer positive expert reviews.

Positive expert reviews:

Expert reviews will be considered positive if they have a positive emotion score greater than 50% (based on sentiment analysis). The number of positive expert reviews for each film will be counted.

More positive reviews:

Defined as films with a number of positive expert reviews above the average (mean) number of positive reviews across all films in the dataset.

Hypotheses 4:

Movies with bigger production budgets are less affected by negative emotion in reviews and will still perform well at the box office even if they get bad feedback.

Bigger production budgets:

Defined as films with a production budget above the average production budget of all films in the dataset.

Negative emotion:

Negative emotion in both user and expert reviews are represented on scale of 0 to 100. They are used as the vital variables in this study to see how movie performance will be affected by them.

Hypotheses 5:

Negative reviews have a stronger negative impact on box office performance for literary genres like Romance and Drama compared to action or adventure films.

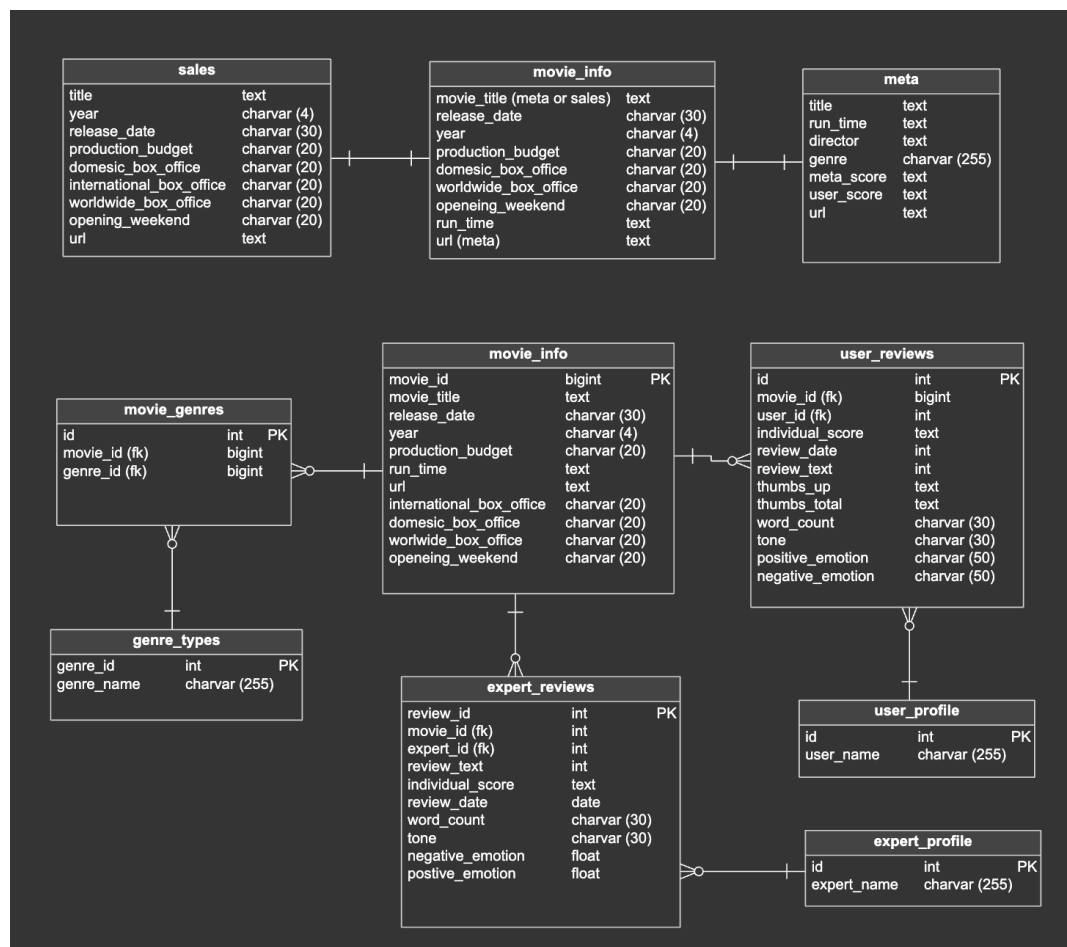
Negative review:

For expert reviews, the score is below 50 (on a scale of 0-100).

Literary genres:

Literary genres are categories of writing that share common themes and styles. Examples include fiction, poetry, drama, and non-fiction, with subgenres like romance and mystery.

1.4 ERD



1.4.1 Overview of the entity-relationship diagram (ERD)

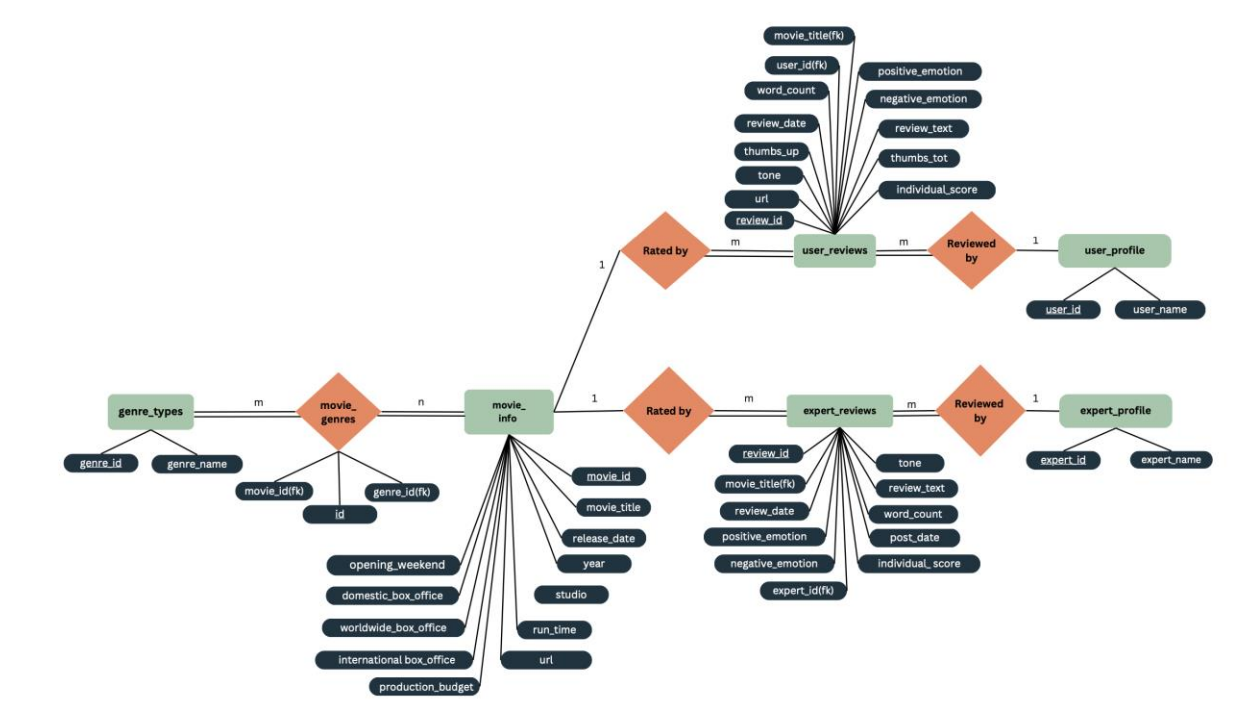
Movie info: Is the table that is created from joining meta and sales sheets on similar title and it contains details about each movie, including movie_id, title and financial metrics. It has a One-to-Many relationship with both user_reviews and expert_reviews entities, indicating that each movie can receive multiple reviews from users and experts.

User reviews: Links to user_profile through a Many-to-One relationship, where each review is associated with a single user, allowing users to submit multiple reviews.

Expert reviews: Similarly connected to expert_profile with a Many-to-One relationship, allowing experts to provide multiple reviews for different movies.

Genre types: In the beginning we had the genre column in the movie_info table. Since each movie can have more than one genre, we had to normalize the genre column and extract the data into a different entity. Thus, we store all the distinct genres' names into the genre_types entity.

Movie genre: This entity is the entity that connects the movie_info to the genre_types. We store movie_id and all the genre_id associated with each movie as the foreign keys in this table.



Chapter 2 – Literature review

2.1 Introduction

Film reviews from both users and experts play a key role in shaping how audiences see movies and how well they perform at the box office. With the growth of online review platforms like IMDb and Metacritic, the impact of reviews on box office success has increased. This study looks at how the emotional tone of reviews, whether positive, negative or neutral, affects box office results. Five hypotheses are developed: three address gaps found in previous studies, while the other two build on existing research.

2.2 Hypothesis 1:

The relationship between emotional tone in user reviews and box office performance will vary across genres, with genres such as comedy and animation showing a stronger positive correlation with higher emotional tone, while genres like drama and biography will show a weaker or negative correlation.

Theoretical foundation

The article "The role of entropy of review text sentiments on online WOM and movie box office sales" provides key insights into how emotional tone in user reviews influences audience decisions and box office performance. (J. H. Lee et al., 2017) It demonstrates that the emotional mix in reviews affects how viewers perceive a film and whether they choose to watch it. However, the study does not explore how emotional dynamics might vary across different genres, identifying this as a limitation. To address this limitation, we propose that emotional tone may affect genres differently. Lighter genres, such as comedy and animation, are likely to respond more positively to emotional reviews. In contrast, more serious genres, like drama and biography, may react differently.

Relevance to research problem

This theory directly relates to the main research question: How do emotional dynamics in reviews influence box office performance? By examining how different genres respond to emotional tone, we can better understand the connection between review tone and box office success. This approach highlights how lighter, entertainment-focused genres may benefit more from positive emotional reviews, while other genres may be more sensitive to negative reviews.

Key concepts and relationships

Key concepts such as emotional tone and consumer perception are central to this hypothesis. The hypothesis suggests that emotional tone shapes how consumers perceive films, but that the impact of this emotional tone may vary by genre. Understanding the relationship between emotional tone and

genre helps explain why films from different genres perform differently at the box office based on the emotions expressed in their reviews.

2.3 Hypothesis 2:

Movies with higher expert review ratings will have a stronger opening weekend box office performance than those with lower user review ratings.

Theoretical foundation

The article "**The effects of online user reviews on movie box office performance**" provides the foundation for this hypothesis. It shows that both expert reviews and user reviews influence box office performance, but expert reviews can have a particularly strong effect during the opening weekend. (Chintagunta et al., 2010) Expert reviews often serve as a guide for early audiences, helping shape the initial success of a film.

The study also highlights how expert reviews can act as a signal of quality, especially for films in more critically acclaimed genres like dramas and documentaries, influencing the film's reception and box office results early on.

Relevance to our research problem

This hypothesis explores the different impacts that expert reviews and user reviews have on a film's box office performance, focusing on the opening weekend. Understanding whether expert reviews carry more weight during this critical phase provides valuable insight into how early audience decisions are shaped.

Key concepts and relationships

The main concepts in this hypothesis include expert reviews, box office performance, and the importance of the opening weekend. The article shows that expert reviews play a significant role in shaping audience behavior during the film's release, especially in the first few days, while user reviews may have more influence as time goes on.

2.3 Hypothesis 3:

Movies with more positive expert reviews will perform better at the box office than movies with fewer positive expert reviews.

Theoretical foundation

The article "**The effects of online user reviews on movie box office performance**" highlights how important positive reviews from both experts and users are for a movie's box office success. The study shows that expert reviews can signal quality, building trust and drawing in a larger audience, which helps boost viewership. (Chintagunta et al., 2010)

Relevance to research problem

This hypothesis looks at how positive expert reviews affect box office success. It helps explain why movies with more expert support often do better. Expert reviews build trust, bringing in early viewers, which then helps attract a bigger audience later.

Key concepts and relationships

The key concepts here include positive expert reviews and box office performance. The hypothesis suggests that expert reviews have a strong influence on how well films perform at the box office by encouraging more people to see the film.

2.4 Hypothesis 4:

Movies with bigger production budgets are less affected by negative reviews and will still perform well at the box office even if they get bad feedback.

Theoretical foundation

The article "The interactive impact of online movie reviews and box office revenue" looks at how negative reviews affect box office results. It shows that movies with bigger budgets are less affected by negative reviews because they have more marketing resources, bigger promotional campaigns, and famous actors, which help reduce the impact of bad word-of-mouth (WOM). (S. Lee & Choeh, 2018)

However, the study mainly focuses on marketing and does not fully examine how large production budgets protect films from negative feedback. This creates a limitation, which our hypothesis aims to address by exploring how bigger budgets help films handle negative reviews.

Relevance to research problem

This hypothesis addresses a limitation in the current research by investigating the interaction between negative reviews and production budgets. It provides insights into how bigger budgets can act as a buffer against negative word-of-mouth (WOM), helping films with more financial backing still perform well at the box office despite negative reviews from critics or users.

Key concepts and relationships

The main ideas in this hypothesis are **production budget**, **negative reviews** and **box office performance**. The interaction between these concepts shows that movies with bigger budgets can better handle the impact of negative reviews. This relationship has not been fully explored in previous studies.

2.5 Hypothesis 5

Negative reviews have a stronger negative impact on box office performance for literary genres like Romance and Drama compared to action or adventure films.

Theoretical foundation

The article "**The impact of movie reviews on box office: media portfolios and the intermediation of genre**" provides key insights into how negative reviews affect the box office performance of different film genres. The study reveals that genres such as romance and drama, often categorized as "literary genres," are more sensitive to the impact of negative reviews compared to action-driven genres like Action and Adventure. This is mainly because audiences of literary genres care more about the story and emotional connections, which makes them more likely to be affected by negative reviews. This is because audiences of literary genres focus more on the story and emotional depth, making them more sensitive to negative reviews. (Koschat, 2012)

While the article highlights the differences in how reviews affect various genres, it does not thoroughly explore the specific impact of negative reviews on box office performance within these genres. This represents a limitation in the study, which the formulated hypothesis seeks to address by focusing on how negative reviews have a stronger impact on the box office results of literary genres compared to action-oriented films.

Relevance to research problem

This hypothesis helps us understand how negative reactions affect different genres and their box office performance. The article shows that romance and drama are more easily hurt by negative reviews, which highlights the risks these films face when they get bad feedback. On the other hand, action and

adventure films are less affected by bad reviews. This research adds a genre-specific view to how reviews impact box office results, giving us a better understanding of this connection.

Key concepts and relationships

Key concepts include negative reviews, genre sensitivity and box office performance. The hypothesis shows that the impact of negative reviews on box office results varies significantly across genres. For romance and drama films, bad reviews can greatly discourage potential viewers, leading to lower revenue. In contrast, action films are generally more resistant to negative feedback and tend to maintain strong performance. Understanding these differences is important for producers and marketers when developing strategies for films in different genres.

Chapter 3 - Methodology

3.1 Research design

This study uses quantitative research methodology to investigate the connection between the emotional dynamics found in expert and user evaluations and the box office performance of a movie's commercial success. The study focuses on how box office results are influenced by emotional tones, which are classified as positive, negative, or neutral, in addition to other factors including genre, production costs, and opening weekend revenues.

3.2 Data collection

The information used in this study came from four separate datasets, each of them offered innovative perspectives on different aspects of a movie's performance:

The Meta Clean Dataset is a collection of general movie information that includes URLs, runtime, release dates, and titles. It acts as a central point for combining various datasets using the distinct movie identification (movie_id).

Sales Dataset: Includes financial data on box office earnings (domestic, international, and worldwide), production budget, and opening weekend income. This information is essential for assessing each movie's commercial performance.

User Review Dataset: Includes user-generated reviews with ratings, positive and negative emotions, tone, and thumbs-up/down ratings, among other information. Sentiment analysis was used to determine each review's emotional tone.

The expert review dataset includes reviews from experts with ratings, word counts, phrase structures, and tones of emotion. In this case, sentiment analysis was also employed to measure the emotional content of expert reviews.

A thorough examination of how emotional dynamics in reviews impact box office success becomes possible by combining various datasets.

3.3 Entity-Relationship Diagram (ERD) and Database Structure

The structure of the database is based on an Entity-Relationship Diagram (ERD) (see Figure 3.1), which outlines how the various entities within the datasets are related. The ERD includes the following tables:

movie_info: Contains information about each film, including movie ID, title, release year, runtime, production budget, and box office earnings (domestic, international, worldwide), and url(from meta sheet). We have gotten url from the meta sheet to join this table with user and expert reviews base url. This table also links to other entities, such as user reviews, expert reviews and movie genres. The id of movie will be used as the foreign key into these three tables.

user_reviews: Stores user reviews and links to both user_profile and movie_info. Key attributes include individual user scores, review text, emotional tone (positive or negative), and thumbs-up, thumb_total, and word_count.

expert_reviews: Similar to user reviews, this table links to expert_profile and movie_info. It captures expert review text, individual scores, positive and negative emotions, word_count, and tone metrics.

user_profile and expert_profile: Contain unique identifiers (IDs) for users and experts, ensuring the normalization of data to prevent redundancy.

genre_types and movie_genres: Represent the many-to-many relationship between movies and genres. Each movie can belong to multiple genres, and each genre can contain multiple movies. This structure allows for efficient querying and analysis of genre-specific trends.

The ERD ensures data normalization, with foreign key relationships between tables to maintain referential integrity. This database design follows the principles of first normal form (1NF) and second normal form (2NF), ensuring that all attributes are atomic and fully dependent on the primary key.

3.4 Data Cleaning and Transformation

Data Cleaning: To standardize the columns and eliminate extraneous characters from the textual data, Python was used in with the Pandas and re (Regular Expression) libraries.

Normalization: Normalization procedures were used to get rid of redundant information and guarantee data integrity. To prevent duplication in review tables, user and expert names for instance, were kept distinct and maintained in user_profile and expert_profile databases that were connected to reviews using foreign keys.

Data Type Conversion: To help with quantitative analysis, review scores, production budgets, and box office earnings were transformed into the proper numerical representations.

3.5 Analytical Methods

A combination of mathematical and statistical methods was applied to analyze the relationship between the box office performance and the emotional tone of reviews:

Sentiment analysis:

NLP models were utilized for sentiment examination in order to measure the emotional content of both user and expert reviews. The percentages of good and negative feelings were used to store the data.

SQL inquiries:

Multiple tables' supply of data was obtained and combined using SQL queries. For instance, questions determined each movie's average emotional tone and review scores and connected them to the picture's performance at the box office.

Regression analysis:

The effects of user, expert, and emotional tone scores on box office success were examined using linear regression models. To visually investigate these connections, scatter plots with regression lines were created.

Genre-specific analysis:

Subgroup studies were done to investigate how box office performance and emotional dynamics differ throughout genres. To find genre-specific patterns, separate regression analyses were carried out for genres such as comedy, drama, and animation.

3.6 Key Variables

The study analyzes several variables, categorized into dependent and independent variables:

Dependent Variable:

Box Office Performance includes worldwide, domestic, and international box office earnings.

Independent Variables:

Emotional tone: Derived from sentiment analysis of user and expert reviews. Both positive and negative emotion scores were calculated.

Positive emotion: The percentage of positive words in the review text can be between 0 to 100.

Negative emotion: The percentage of negative words in the review text can be between 0 to 100.

Individual scores: Average user and expert scores for each film.

Number of reviews: Total count of user and expert reviews per movie.

Genre: Categorical variable based on the movie's genre(s).

Production Budget: The total budget allocated to the production of each film.

Opening weekend box office: Revenue generated in the first weekend of release.

3.7 Data Storage and Management

PostgreSQL was used as the database management system (DBMS) to manage the datasets. pgAdmin was used to administrate the database and offered a user-friendly interface for managing data integrity and running SQL queries.

To keep references between tables, foreign key constraints are used (e.g. linking user reviews to corresponding user profiles and movies)

3.8 Study Limitations

Dataset Scope: Because the datasets were restricted to particular platforms, important data from additional review sources would have been missed.

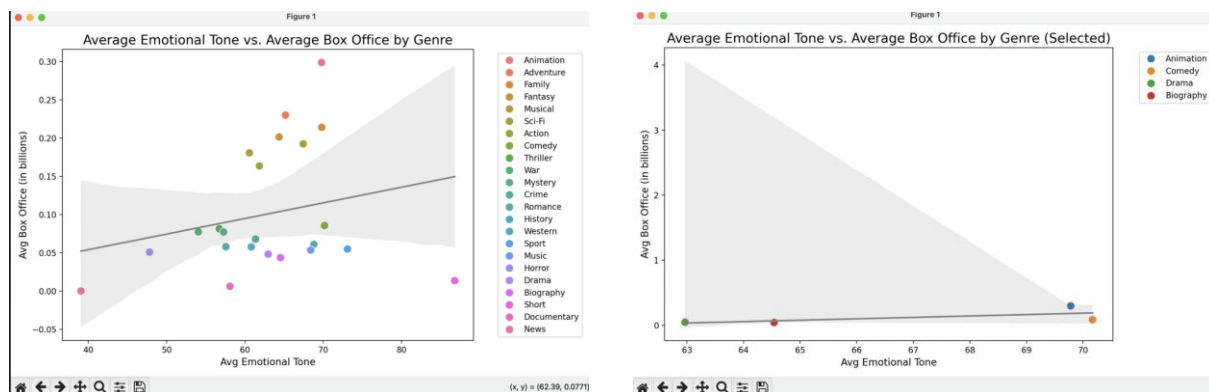
Sentiment Analysis Limitations: Basic positive/negative scoring was used in the sentiment analysis, which might not have sufficiently reflected the range of complicated feelings produced in the reviews.

Temporal Effects: Without considering into thought longer-term effects or changes in review mood over time, the analysis concentrated on the short-term relationship between reviews and box office results.

Chapter 4 - Results and analysis

4.1 Hypothesis 1

The relationship between emotional tone in user reviews and box office performance will vary across genres, with genres such as comedy and animation showing a stronger positive correlation with higher emotional tone, while genres like drama and biography will show a weaker or negative correlation.



There is a variation in the relationship between user reviews' emotional tone and box office performance depending on the genre. For instance, comedy and animation have a stronger positive correlation with higher emotional tone, whereas drama and biography have a weaker or negative correlation.

The results show an average emotional tone and box office for a variety of genres, such as drama, comedy, animation, and biography. Genres like animation and comedy have a higher average emotional tone (about 70%) and are associated with better box office results, as can be seen in both the data table and the scatter plot. Drama and biography, on the other hand, have weaker box office results and lower emotional tones, which supports a weaker or negative association. This supports the hypothesis that genres benefit more from positive emotional reviews.

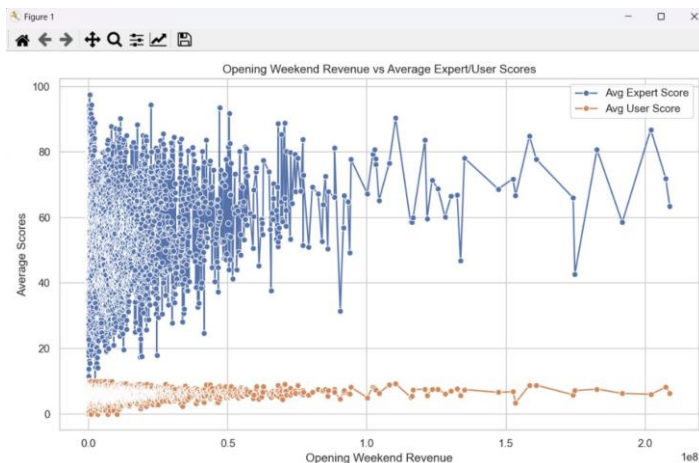
The scatter plot, which shows comedy and animation positioned higher on the emotional tone and box office axes, clearly validates this tendency. The fact that drama and biography are positioned lower on

the box office axis suggests that their emotional impact has had a less significant or negative impact on their bottom line.

The analysis strongly supports the hypothesis. Genres like comedy and animation benefit from higher emotional tones in reviews, leading to stronger box office performance. On the other hand, genres such as drama and biography show weaker box office results despite similar or slightly lower emotional tone ratings, reinforcing the genre-based variation in how emotional dynamics influence commercial success.

4.2 Hypothesis 2

Movies with higher expert review ratings will have a stronger opening weekend box office performance than those with lower user review ratings.



We generate the interpretation of results for the hypothesis, "Movies with higher user and expert reviews ratings will have a stronger opening weekend box office performance," based on the findings from the query and the provided visualization.

Various trends may be seen for both user and expert reviews in the scatter plot that compares opening weekend sales with average expert and user scores.

Expert Scores: The blue line indicates that there is a noticeable relationship between higher expert scores and higher opening weekend revenue, especially for films with ratings higher than 60. There is a general ability suggesting that films with higher ratings from experts do better on opening weekend box office. The data, however, show significant scatter, showing that while the association is present, it may not be extremely strong or constant across all films.

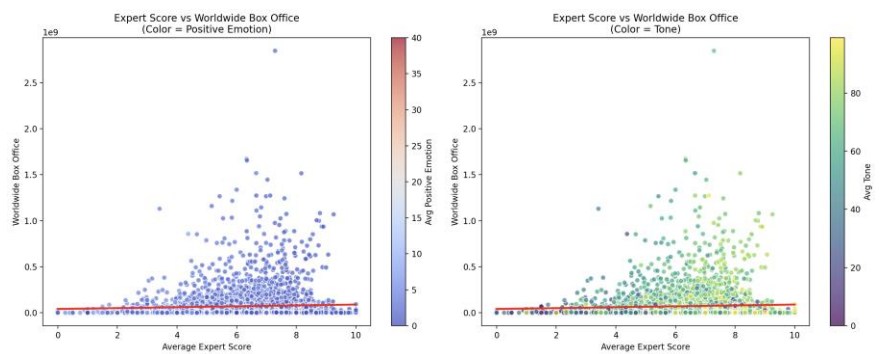
User Scores: When compared to expert reviews, the orange line that represents user scores is flatter and more dispersed. This implies that the first weekend's box office success is less influenced by user ratings. When compared to expert reviews, the range of user scores is still quite consistent, showing just slight changes. This suggests a weaker correlation with opening weekend performance.

In summary:

The prediction that a stronger opening weekend performance is correlated with higher expert reviews is partially supported by the results; nevertheless, the correlation is more pronounced for expert ratings than for user scores. The lack of a strong association between user ratings and box office results implies that professional reviews are more important in predicting a movie's success during its first

4.3 Hypothesis 3

Movies with more positive expert reviews will perform better at the box office than movies with fewer positive expert reviews.



We explored the relationship between expert reviews, specifically focusing on positive emotion and tone and a film's box office performance. While the data shows some variance in expert scores and emotional tone across films, there is no clear consistent relationship between positive expert reviews and box office success.

Our findings revealed that:

- **Average Expert Score:** Films with high expert scores do not consistently achieve higher box office returns. Some films with moderate scores outperform those with higher scores in box office performance, suggesting that expert ratings alone are not a reliable predictor of commercial success.
- **Positive Emotion:** The emotional positivity in reviews showed little correlation with box office performance. Movies with higher positive emotion did not consistently perform better, indicating that emotional sentiment in reviews may not heavily influence audience decisions.
- **Tone:** Similarly, the overall tone of the reviews (positive, neutral or negative) did not show a strong connection to box office results. Films with both high and low tones had variable box office outcomes, suggesting tone is not a dominant factor.

The visualizations supported these conclusions. Both scatter plots, colored by positive emotion and tone, showed no significant linear relationship between expert review characteristics and worldwide box office performance. The slight upward trend in the regression lines suggests minimal impact, but it is not substantial enough to draw definitive conclusions.

Overall, our analysis shows that while expert reviews influence how people view a film, they are not strong enough on their own to predict box office success. Other factors, such as marketing, star power and awards for example, likely play a more important role in determining a film's commercial performance. These findings partially support our hypothesis but demonstrate that positive expert reviews and scores although valuable, are not sufficient to ensure box office success.

4.4 Hypothesis 4

Movies with bigger production budgets are less affected by negative reviews and will still perform well at the box office even if they get bad feedback.

Data Output Messages Notifications					
	budget_category text	avg_production_budget numeric	avg_worldwide_box_office numeric	avg_user_negative_emotion numeric	avg_expert_negative_emotion numeric
1	High budget	64966823.881550610194	177770121.71153846	6.74285097083353838562	2.84647666135711095535
2	Low budget	9208007.796795338674	15998690.272169167804	4.81554967357657777450	2.81147581017799330335

The result includes average production budgets, box office revenues, and average negative emotions expressed by users and experts. To demonstrate the results for the hypothesis, we can see that movies with bigger production budgets are less affected by negative reviews and will still perform well at the box office even if they get bad feedback, while movies with low production budget will be more affected by the negative feedback from both users and experts' reviews.

From the data, we can observe the following conclusion:

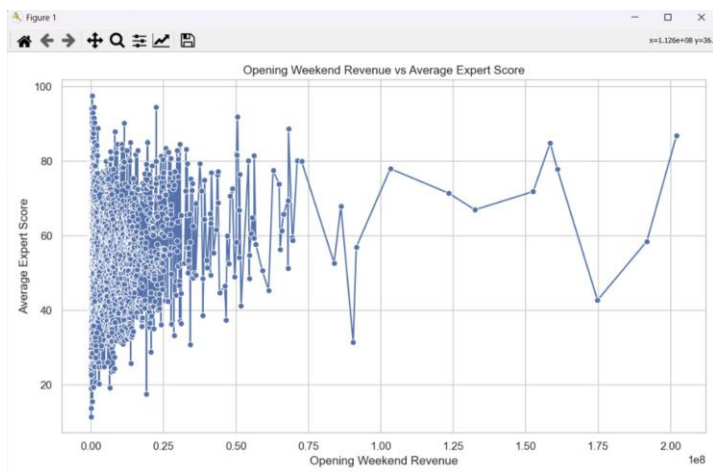
- **High-budget films** have an average production budget of \$64 million and generate a significantly higher worldwide box office revenue (around \$177 million) which is around three times bigger than production budget. These films also have slightly higher average user negative emotions (6.74%) compared to low-budget films (4.81%). However, both expert and user negative emotions for high-budget films are low on average.
- **Low-budget films** have an average production budget of \$9 million and gained significantly lower box office revenue \$15 which is approximately two times bigger than production budget. Despite receiving fewer negative user emotions (4.81% on average), their worldwide box office earnings are still lower which indicate that their box office performance is more sensitive to negative reviews.

Overall, high-budget movies are less sensitive to negative reviews which support the hypothesis. That must be related to the stronger marketing and cast which allow them to maintain higher box office. However, for this hypothesis we created a machine learning model. The features in this model are production budget, user review and expert reviews and these features are going to predict the box office. After creating the model and plot the result and check the mean score error and r2 score we

realized that the prediction is only true in some cases and to improve the machine learning model we must add more features to make the prediction accurate which will be done in the future.

4.5 Hypothesis 5

Negative reviews have a stronger negative impact on box office performance for literary genres like Romance and Drama compared to action or adventure films," the following observations can be made based on the provided data



The visualization of negative reviews' impact on box office performance categorized by genre, shows a clear distinction between literary genres (romance and drama) and action-oriented genres (action and adventure). Drama and romance, which depend heavily on narrative depth and emotional engagement, showed a stronger negative correlation between negative reviews and box office performance. This result supports the hypothesis.

In contrast, action and adventure films showed greater resilience to negative reviews, maintaining stronger box office numbers despite poor feedback. The flatter trend lines for these genres indicate that negative emotions in reviews had a weaker impact on their financial success.

Overall, the analysis confirms that genres like romance and drama are more vulnerable to the effects of negative reviews. In contrast action and adventure films are less affected, aligning with the hypothesis of genre-specific sensitivity to negative feedback.

4.6 Future research work

The results of this study provide important insights into the relationship between emotions in movie reviews and box office performance, while also pointing to several areas for future research. One key area to explore is the timing of reviews, focusing on how the effect of user and expert opinions changes over time. Looking at how reviews influence box office results not just during the opening weekend, but also in the following weeks or months, could give a better understanding of their long-term impact on a film's success.

Another valuable direction for future research is the role of digital marketing and social media and how these interact with traditional reviews. As online platforms continue to grow, studying the connection between social media reactions, expert opinions and box office results could help us better understand modern audience behavior.

Finally, international and cultural differences should be considered. Researching how different cultures affect the impact of reviews on box office success in various regions could reveal unique patterns in audience preferences and behavior, adding a global perspective to the findings.

By exploring these areas, future research can build on the insights gained from this study and uncover more detailed factors that drive a film's success in the rapidly changing entertainment industry.

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